

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
Second Semester B.Tech Degree Examination June 2022 (2019 scheme)

Course Code: EST130
Course Name: BASICS OF ELECTRICAL AND ELECTRONICS ENGINEERING
(2019 -Scheme)

PART 2 : BASIC ELECTRONICS ENGINEERING

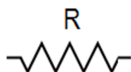
PART A

Answer all questions, each carries 4 marks Marks

12. Draw the symbol of resistor and explain any three specifications. (4)

Answer

The symbol for Resistor is shown below.



Specifications:

1. Nominal Resistance : The nominal resistance is the resistance that can be expected when purchasing a resistor. They do not come in all values as per the need of designers.
2. Tolerance : The tolerance of the resistor is the amount by which the resistance of the resistor may vary from its stated value.
Larger the tolerance – More the resistance variation.
Smaller the tolerance – More stable resistance.
Common way of specifying the resistor tolerance is in PERCENTAGE - %
3. Temperature Co efficient (TC): TC describes the relative change of resistors physical property that is associated with the given change in temperature.
PTC – Temperature increases , Resistance also increases.
NTC – Temperature increases , Resistance decreases.

13. For an NPN Transistor, $\alpha = 0.98$, $I_B = 100\mu A$, Find I_E and I_C . (4)

Answer

According to the relation between Common emitter current gain (β factor) and Common base current gain (α - factor) equation;

$$\beta = \frac{\alpha}{1 - \alpha}$$

$$\beta = \frac{0.98}{1 - 0.98} = 49$$

We have the equation for Common emitter current gain (β factor) as;

$$\beta = \frac{I_C}{I_B}$$

$$\text{Therefore, } I_C = \beta I_B = 49 * 100\mu A = 4.9mA$$

The relation between Collector current, Emitter current and Base current is given as;

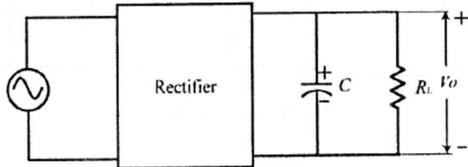
$$I_E = I_B + I_C$$

$$\text{So } I_E = 100\mu A + 4.9mA = 5mA$$

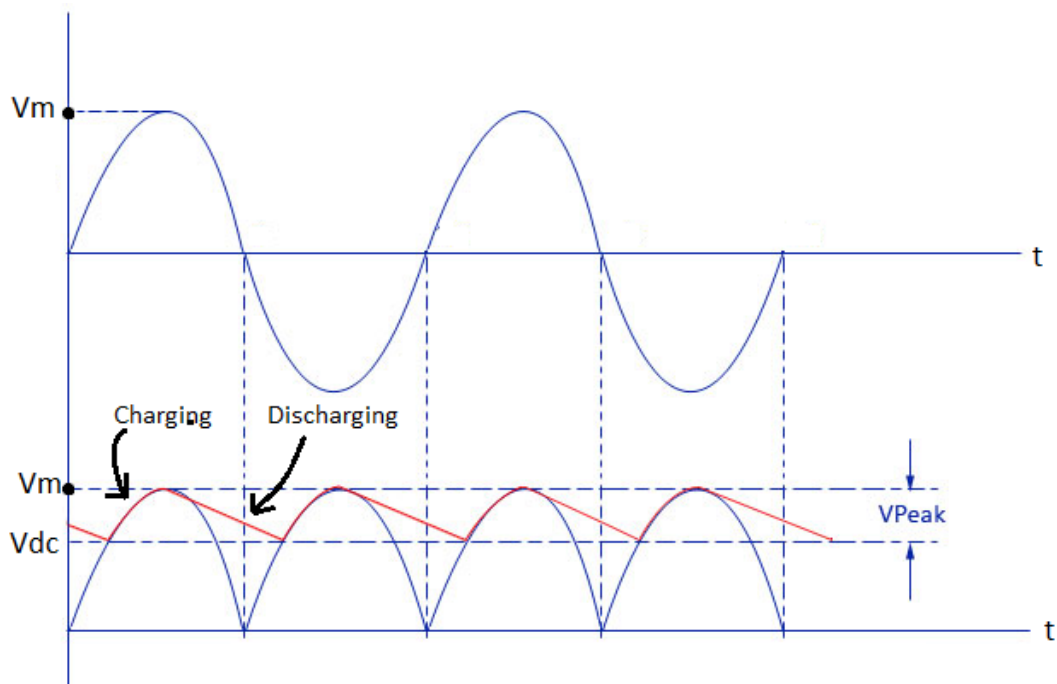
14. Explain the action of the shunt capacitor filter. (4)

Answer

The basic diagram for a shunt capacitor filter for DC power supply is shown below.



The filter circuit is necessary for smoothing of the voltage obtained by the rectifier. The obtained DC voltage contains AC components. These AC components are called ripples. The filter circuit is needed to remove the ripples from DC output voltage. The capacitor has the basic property of opposing the dc components and allows the ac components of a signal to pass through it. Capacitor is connected between the load and output of the rectifier circuit. When the rectifier output voltage is increasing to the maximum voltage (V_m), the capacitor charges to V_m . When the rectifier output is decreased from maximum voltage V_m , the capacitor starts to discharge through the load resistor R_L . Ripples can be reduced by allowing the capacitor to discharge slowly, So the value of load resistor can be taken of large value.

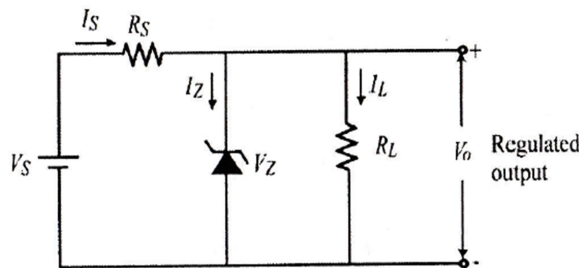


15. Explain the working principle of Zener voltage regulator. (4)

Answer

A zener diode provides constant voltage when operated in the reverse breakdown region irrespective of current through it and variation in the applied voltage. Thus a Zener diode can be used as a voltage regulator to provide constant dc output voltage.

The circuit diagram of a simple Zener voltage regulator is shown below.



The R_S is known as series resistance that limits the current flow. It is essential that the input voltage V_S should be greater than zener voltage regulator and the current flow through the zener should be in between the minimum and maximum zener currents.

Line Regulation of zener diode regulator - with varying input voltage

$$I_S = I_Z + I_L$$

Here the load resistance R_L is fixed and input voltage V_S is varied within limits.

Assume the input voltage V_S increases, the input current (I_S) also increases. This raises the current through the zener diode (I_Z), without affecting the load current I_L . The increase in input current will also increase the voltage drop across the series resistance R_S , thereby keeping the output voltage V_o across load resistor R_L as constant. If the input voltage is reduced, the input current also decreases. Then the current through the zener diode will also decrease. The voltage drop across the series resistance will decrease. Thus the load voltage and load current remains constant.

Line Regulation determines the amount the load voltage changes (ΔV_o) when the input source voltage changes (ΔV_S). In ideal case, line regulation is zero.

$$\text{Percentage line regulation} = \frac{\Delta V_o}{\Delta V_S} \times 100\%.$$

Load Regulation - with varying load current

Here the input voltage V_S is fixed and load current I_L is varied by varying the load resistance R_L .

We know, $I_S = I_Z + I_L$

When load current increases, current through the zener diode will decrease as source current remains constant. And vice versa when load current decreases.

16. Differentiate between amplitude modulation (AM) and frequency modulation (FM). (4)

Answer

Amplitude modulation is the process of varying the amplitude of the carrier wave in accordance with the instantaneous values of the message signal (or modulating signal or baseband signal)

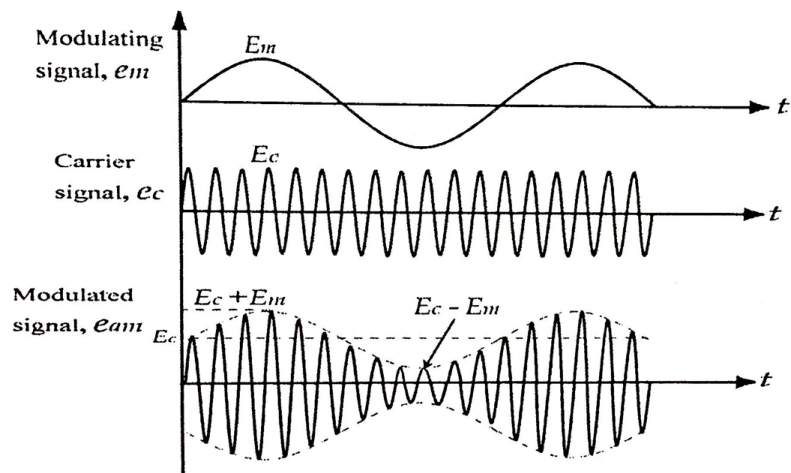


Figure 11.2: Amplitude modulation

Frequency Modulation is process of varying the frequency of the carrier signal in accordance with the instantaneous amplitude of the modulating signal

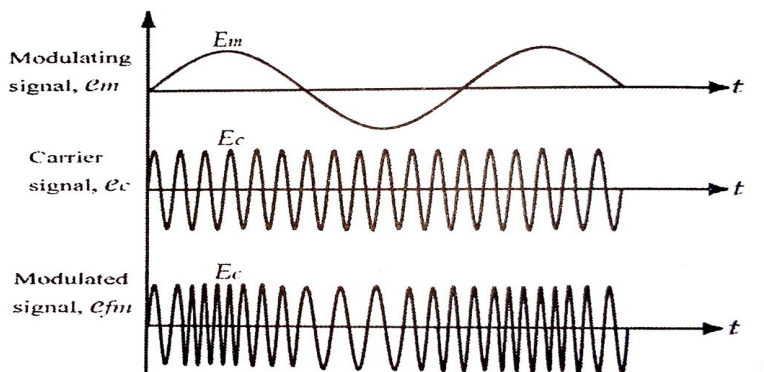


Figure 11.4: Frequency modulation

AM	FM
1. Amplitude of the carrier is varied 2. Suitable for long distance communication 3. Poor Noise immunity 4. Less band width requirement as compared to FM 5. More power is required for transmission 6. Power wastage is high in AM 7. Circuitry is simple 8. Co channel interference is more 9. Line of sight is not required	1. Frequency of the carrier is varied 2. Suitable for short distance communication. 3. Better noise immunity as compared to AM 4. More band width is required for FM signal 5. Power requirement is less as compared to AM 6. No power is wasted 7. Circuits required for FM is complex and expensive 8. Co channel interference is high 9. Line of sight communication

PART B

Answer one full question from each module, each question carries 10 marks.

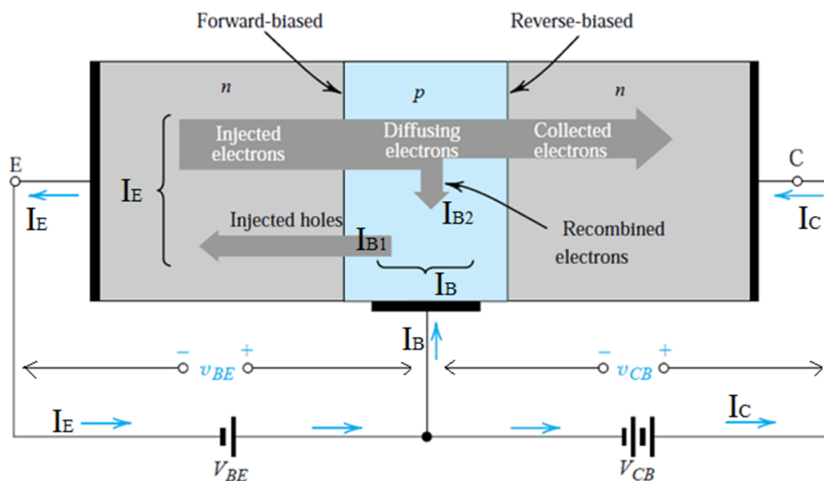
MODULE 4

17. a) Explain with necessary diagrams, the principle of operation of NPN transistor (5)

Answer

Let's consider the active mode of operation. Emitter-Base junction is forward biased and Collector-Base Junction is reverse biased as shown below. The Voltage applied at the emitter base junction is denoted as V_{BE} and that of Collector base junction is denoted as V_{CB} .

The value of $V_{CB} \gg V_{BE}$



The voltage V_{BE} will cause the electrons in the Emitter (n-type) to move towards the Base (p-type). These are known as Injected electrons and Holes from base region to move towards Emitter, which are known as injected holes. The flow of electrons in the base is due to Diffusion. Since base is a p-type material which has holes in it, some of the injected electrons will Recombine with these holes. But due to the thin size, small amount of doping of base and the large reverse bias voltage (V_{CB}) at the collector, most of the electron will move towards the collector without recombining with the hole. These electrons which are moved to the collector will be collected by the collector terminal.

Total Current flow in the transistor will be:

- Due to the flow of electrons from Emitter to Base (I_{E1}) and flow of holes from Base to Emitter (I_{B1}). Both these currents will be flowing from Base to Emitter. This current is known as Emitter current and is denoted as $I_E = I_{E1} + I_{B1}$
- Due to the recombination of electrons and holes in the base (I_{B2}) and flow of holes from Base to Emitter (I_{B1}). Both these currents together are known as Base current and denoted as I_B . This current is effectively flowing from base to emitter.
- Due to the flow of electrons in the Collector region. This current is known as collector current which is flowing from collector towards base and is denoted as I_C

Therefore the total current in transistor can be written as: $I_E = I_B + I_C$

b) Describe the color coding of a resistor with an example. (5)

Answer

The resistance values of resistors are denoted using color coding.

Color	Digit	Multiplier	Tolerance (%)
Black	0	10^0 (1)	
Brown	1	10^1	1
Red	2	10^2	2
Orange	3	10^3	
Yellow	4	10^4	
Green	5	10^5	0.5
Blue	6	10^6	0.25
Violet	7	10^7	0.1
Grey	8	10^8	
White	9	10^9	
Gold		10^{-1}	5
Silver		10^{-2}	10
(none)			20

There are two methods of colour coding.

Four band Color coding: In this method, there will be four colour bands on the resistor. The First two bands represent Digits, Third band represents Multiplier (10^x) and Fourth band for Tolerance.

Five band Color coding: In this method, there are five colour bands on the resistor. First three bands for Digits, Fourth band represents Multiplier (10^x) and Fifth band for Tolerance.

Eg. Consider a resistor with colour bands Yellow, Violet, Red and Gold

The resistance value (based on four colour band method) can be calculated as:

Yellow: 4

Violet: 7

Red: 10^2

Gold: $\pm 5\%$

Resistance value: $47 * 10^2 \pm 5\% \Omega = 4.7 \pm 5\% k\Omega$

OR

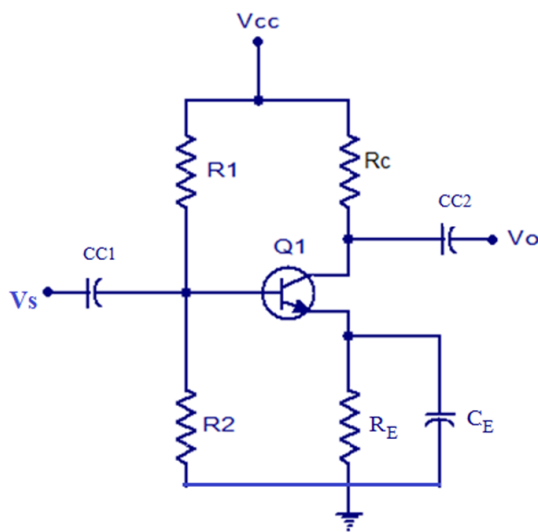
18. a) Draw the circuit diagram of a common emitter amplifier. (3)

Answer

The circuit diagram is shown below. The dc source V_{cc} forward bias the emitter junction and reverse bias the collector junction. After the transistor has been biased with a Q-point near the middle of the load line, AC signal can be applied to the base. The ac input is applied between base & emitter terminals and output

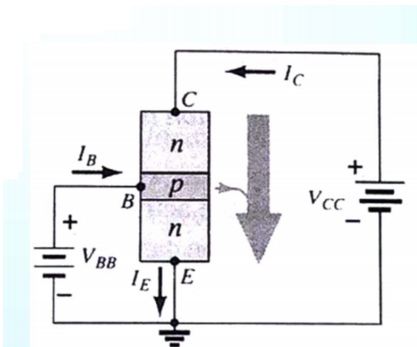
is taken from collector & emitter terminals. The resistor R_{B1} and R_{B2} are acting as a voltage divider circuit and voltage across R_2 ensures that the Base emitter junction is forward biased. The resistance R_C is connected to the collector terminal which is acting as a load.

The capacitor $CC1$ (Coupling capacitor) blocks the dc components from the source and passes only ac components to the input and the coupling capacitor $CC2$ blocks the dc components and allows the ac to the output terminal. Capacitor C_E is known as Bypass capacitor, which ensures the stability of Q-point by providing a path to the ac emitter current.



b) Explain the input and output characteristics of common emitter configuration with neat diagrams (7)

The circuit representation for common emitter configuration is shown below:

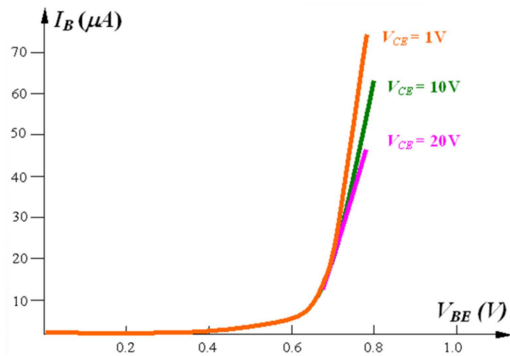


In common emitter configuration, the curve drawn between base current I_B and base-to-emitter voltage V_{BE} for a given value of collector-emitter voltage V_{CE} gives the input characteristics.

Input voltage V_{BE} is applied between base and emitter terminals and output voltage V_{CE} is applied across emitter and collector. The output current I_C is taken across the emitter and collector terminals. The input side is forward biased and the output side is reverse biased.

Emitter base region acts like forward biased diode and so the depletion region is very small. Emitter collector region acts like a reverse biased diode and the depletion region is large. The input current I_B is measured in μA because the base region is very lightly doped.

It is observed that when the input voltage V_{BE} is increased initially there is no current produced, further when it is increased the input current I_B increases steeply. When the output voltage V_{CE} is further increased the curve shifts right side.

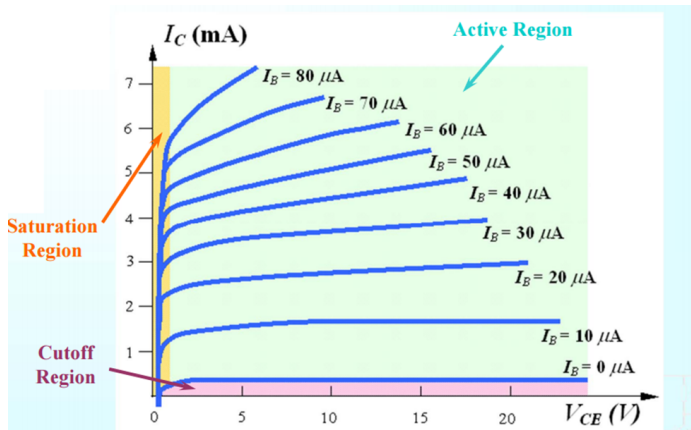


Cut in voltage or Knee voltage: The voltage at which the base current starts to increase. It is 0.7v for silicon & 0.3v for Germanium. It can be observed that as the collector-to-emitter voltage (V_{CE}) is increased, the curve shifts downwards due to increase in depletion width of base region.

Dynamic input resistance : The reciprocal slope of the characteristics is the ratio of change in base-to-emitter voltage to the resulting change in base current for a constant collector to emitter voltage is known as dynamic resistance.

$$\text{AC or Dynamic input resistance} = \frac{\Delta V_{BE}}{\Delta I_B} \text{ for } V_{CE} \text{ constant}$$

The output characteristic relates the output current (I_C) to the output voltage (V_{CE}) for various levels of input current (I_B). In active region when the output voltage is increased there is very slight change in the output voltage. The curve looks almost flat in the active region. Cut off region is the region where the input current is below zero. When both the junctions are forward biased, it is in saturation region.



Output resistance : Ratio of a change in collector to emitter voltage to the resulting change in collector current for a constant base current.

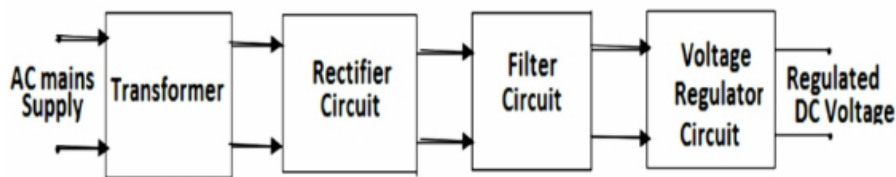
$$\text{Output resistance, } r_o = \frac{\Delta V_{CE}}{\Delta I_C} \text{ for } I_B \text{ constant.}$$

MODULE 5

19) Describe the components of a DC power supply using a neat block diagram. (10)

The basic power supply is constituted by four elements viz a transformer, a rectifier, a filter, and a regulator put together. The output of the dc power supply is used to provide a constant dc voltage across the load. Transformer is used to step-up or step-down (usually to step-down) the supply voltage as per need of the solid-state electronic devices and circuits to be supplied by the dc power supply. It can provide isolation from the supply line-an important safety consideration. It may also include internal shielding to prevent unwanted electrical noise signal on the power line from getting into the power supply and possibly disturbing the load.

Rectifier is a device which converts the sinusoidal ac voltage into either positive or negative pulsating dc. P-N junction diode, which conducts when forward biased and practically does not conduct when reverse biased, can be used for rectification i.e. for conversion of ac into dc. The rectifier typically needs one, two or four diodes. Rectifiers may be either [half-wave rectifiers](#) or full-wave rectifiers ([centre-tap](#) or [bridge](#)) type.



Block Diagram of DC Regulated Power Supply

The output voltage from a rectifier circuit has a pulsating character i.e., it contains unwanted ac components (components of supply frequency f and its harmonics) along with dc component. For most supply purposes, constant direct voltage is required than that furnished by a rectifier. To reduce ac components from the rectifier output voltage a filter circuit is required.

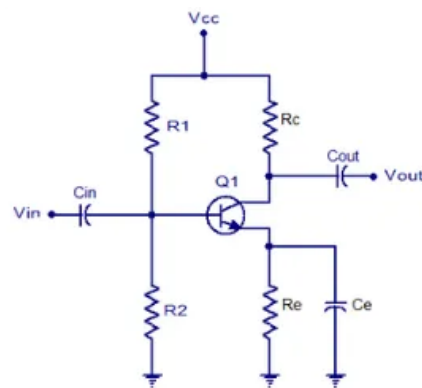
Thus filter is a device which passes dc component to the load and blocks I ac components of the rectifier output. The magnitude of output dc voltage may vary with the variation of either the input ac voltage or the magnitude of load current. So at the output of a rectifier filter combination a voltage regulator is required, to provide an almost constant dc voltage at the output of the regulator.

OR

20) Explain the working of RC coupled amplifier with circuit diagram and relevant waveforms. Also explain the frequency response of RC coupled amplifier. (10)

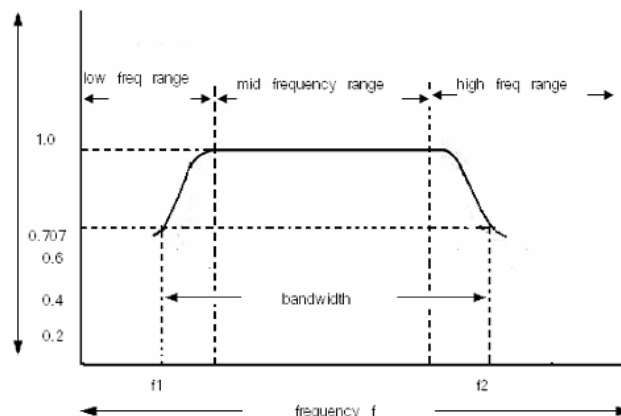
A **Resistance Capacitance (RC) Coupled Amplifier** is basically a multi-stage amplifier circuit extensively used in electronic circuits. Here the individual stages of the amplifier are connected together using a resistor–capacitor combination due to which it bears its name as RC Coupled. the design of individual stages of the **RC coupled amplifiers** is similar to that in the case of common emitter amplifiers in which the resistors R_1 and R_2 form the biasing network while the emitter resistor R_E form the stabilization network.

Here the CE is also called bypass capacitor which passes only AC while restricting DC, which causes only DC voltage to drop across R_E while the entire AC voltage will be coupled to the next stage.



Further, the coupling capacitor C_C also increases the stability of the network as it blocks the DC while offers a low resistance path to the AC signals, thereby preventing the DC bias conditions of one stage affecting the other. In addition, in this circuit, the voltage drop across the collector-emitter terminal is chosen to be 50% of the supply voltage V_{CC} in order to ensure appropriate biasing point.

Frequency response of Common Emitter RC Coupled Amplifier is shown below:



In mid frequency range(50 Hz to 20KHz),the voltage gain of the amplifier is constant, as is from the analysis. With the increase in the frequency in this range, the reactance of the coupling capacitor C_c reduces thereby increasing the gain but at the same time lower capacitive reactance causes higher loading resulting in lower voltage gain. Thus the two effects cancel each other and uniform gain is obtained in mid frequency range.

At low frequency range(below 50 Hz), higher capacitive reactance of coupling capacitor C_c allows very small part of signal to pass from one stage to the next and also because of higher reactance of emitter bypass capacitor C_E , allows very small part of signal to pass from one stage to the next and also because of higher reactance of emitter bypass capacitor C_E , the emitter resistor R_E is not effectively shunted. Thus the voltage gain falls off at low frequencies.

MODULE 6

21. a) Explain the concept of cells in cellular communication. (3)

Cellular communication is based on the geographic division of the communication coverage area into cells, and within cells. Each cell is allocated a given number of frequencies (or channels) that allow a large number of subscribers to conduct conversations simultaneously.

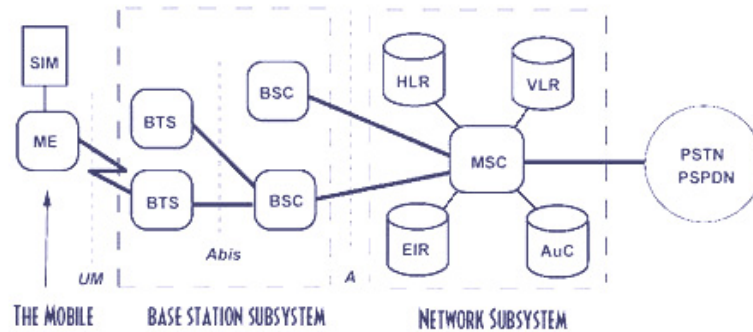
The common element of all generations of cellular communication technologies is the use of defined radiofrequencies (RF), as well as frequency reuse. This enables the provision of a service to a large number of subscribers while reducing the number of channels (band width). It also enables the creation of wide communication networks by fully integrating the advanced capabilities of the mobile phone. The increase in demand and consumption, as well as the development of different types of services, accelerated the rapid technological development of advanced cellular communication networks, together with unceasing improvement of the cellular devices themselves.

b) Draw the block diagram of GSM and explain the principle of operation. (7)

Fig 1 shows GSM architecture. The network mainly consist of User Equipment (UE), Base transceiver station (BTS), Mobile switching center (MSC). The GSM contains most of the necessary capabilities to support packet transmission over GSM. The critical part in the GPRS network is the mobile to GSN (MS-SGSN) link which includes the MS-BTS, BTS-BSC, BSC-SGSN, and the SGSN-GGSN link .fig 1 shows block diagram of GSM architecture.

User Equipment (UE)—These are the users .Number of users are controlled by one BTS

Schematic of a GSM system.



1. The mobile stations (MS) communicate with the base station subsystem over the radio the radio interface.
2. The BSS called as radio the subsystem, provides and manages the radio transmission path between the mobile stations and the Mobile Switching Centre(MSC).It also manages radio interface between the mobile stations and other subsystems of GSM.
3. Each BSS comprises many Base Station Controllers(BSC) that connect the mobile station to the network and switching subsystem (NSS) through the mobile switching center
4. The NSS controls the switching functions of the GSM system.It allows the mobile switching center to communicate with networks like PSTN, ISDN, CSPDN, PSPDN and other data networks.
5. The operation support system (OSS) allows the operation and mantanance of the GSM system. It allows the system engineers to diagnose,troubleshoot and observe the parameters of the GSM systems.The OSS subsystem interacts with the other subsystems and is provided for the GSM operating company staff that provides service facilities for the network.

Base station(BSS)-- The following stations subsystem comprises of two parts:

1. Base Transceiver Station (BTS).
2. Base Station Controller(BSC).

The BSS consists many BSC that connect to a single MSC. Each BSC controls upto several hundred BTS.

Base Transreciver Station(BTS)-BTS

It has radio transreciever that define a cell and are capable of handelling radio link protocols with MS.

Functions of BTS are

1. Handelling radio link protocols
2. Providing FD communication to MS.
3. Interliving and de- interliving.

Base station controller(BSC) IT manages radio resources for one or more BTS.It controls several hundred BTS al are connected to single MSC.

Functions of BTS are

- To control BTS.
- Radio resource management
- Handoff management and control
- Radio channel setup and frequency hoping

Network subsystem(NSS)

- 1.It handels the switching of GSM calls between external networks and indoor BSC
- 2.It includes three different data bases for mobility management as

- A .HLR (Home Location Register)
- B .VLR (Visitor Location Register)
- C. AUC (Authentication center)

Mobile switching center (MSC)--

It connects fix networks like ISDN ,PSTN etc.

Following are the functions of MSC

1. Call setup, supervision and relies
2. COLLECTION OF BILLING INFORMATION
3. Call handelling / routing
4. Management of signalling protocol
5. Record of VLR and HLR

OR

22. a) Describe the principle and working of an antenna. (6)

Antennas are devices that interact with radio waves travelling through space from one point to another. These devices have been around for a long time and are used in a plethora of electronic systems requiring the transmission of the signal via electromagnetic waves such as radar systems, radio, and television.

An antenna consists of a metal conductor that conveys radio frequency (RF) waves between two points in space. This device can either transmit a signal or receive one. When a voltage is applied to a transmitting antenna, it generates radio signals which travel to a receiving antenna where the signal is converted back into electrical energy in the form of information.

There are certain fundamental properties all antennas possess which enable them to function effectively. These [key properties of antennas](#) are listed below:

- Intensity of radiation
- Radiation pattern
- Radiation efficiency and power gain
- Input impedance
- Effective length
- Bandwidth
- Effective aperture
- Polarization

Intensity of radiation

The intensity of radiation of a given antenna refers to the power per unit angle measured in Watts per steradian (W/Sr)

Radiation Pattern

In most instances, an antenna cannot radiate equal amounts of energy in all directions. Most antennas emit radio waves maximally in one direction while the energy emitted in other areas is negligible. The radiation pattern of an antenna is represented using the quantity Field Strength. Field strength is measured at a distance from the antenna in volts/meter

Radiation Efficiency and Power Gain

How efficiently an antenna functions are limited by the conductor material it is composed of. As aerials are made of materials with imperfect conductivity, their efficiency is not a hundred per cent. The radiation efficiency of an antenna is the ratio of radiated power to input power. Power gain refers to the power radiated in a particular direction when compared to the total power input.

b) What is frequency reuse? Explain with a diagram. (4)

Frequency Reuse is the scheme in which allocation and reuse of channels throughout a coverage region is done. Each cellular base station is allocated a group of radio channels or Frequency sub-bands to be used within a small geographic area known as a cell. The shape of the cell is Hexagonal. The process of selecting and allocating the frequency sub-bands for all of the cellular base station within a system is called **Frequency reuse** or **Frequency Planning**.

Silent Features of using Frequency Reuse:

- Frequency reuse improve the spectral efficiency and signal Quality (QoS).
- Frequency reuse classical scheme proposed for GSM systems offers a protection against interference.
- The number of times a frequency can be reused is depend on the tolerance capacity of the radio channel from the nearby transmitter that is using the same frequencies.
- In Frequency Reuse scheme, total bandwidth is divided into different sub-bands that are used by cells.
- Frequency reuse scheme allow WiMax system operators to reuse the same frequencies at different cell sites.